



Task 1 – Credit Submission

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A. Task Credit1a

First and foremost, students are required to download the database file **customer(1)-1.accdb** from the ZIP file named **CUSTOMERDB [ACCDB, 1.4MB]** available on Canvas and rename the **CUSTOMER** table by appending the last four digits of their student number, resulting in the table name **CUSTOMER5520**.

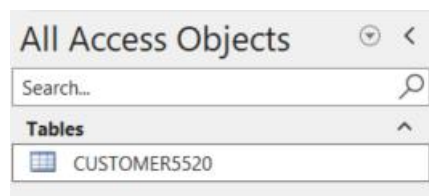


Figure 1: Renamed CUSTOMER Table with Student ID Suffix (5520)

Next, I selected the Create tab and chose Query Design to create a query named **TICA_5520**. The query includes the following columns: Customer ID, Surname, Gender, Loyalty Points, Rating Code, and State, which match the assignment requirements.

- The State must be Tasmania
- The Loyalty Points must be greater than or equal to 7500
- The Gender must be M
- The Rating code must be 1 or 3

Field:	Custid	Surname	Gender	LoyaltyPts	Rating	State
Table:	CUSTOMER5520	CUSTOMER5520	CUSTOMER5520	CUSTOMER5520	CUSTOMER5520	CUSTOMER5520
Sort:						
Show:	☒	☒	☒	☒	☒	☒
Criteria:			"M"	>=7500	1 Or 3	"Tasmania"
or:						

Figure 2: Query Design Grid for TICA_5520

CustId	Surname	Gender	LoyaltyPts	Rating	State
1024	Fidler	M	8380	1	Tasmania
1032	McMaster	M	8870	1	Tasmania
1043	Ferguson	M	7500	3	Tasmania
1058	Trumper	M	8690	1	Tasmania
1067	Whish	M	8210	3	Tasmania
1076	Espie	M	9130	1	Tasmania
1078	Minifie	M	9210	1	Tasmania
1092	McIlwraith	M	8760	3	Tasmania
1104	Persse	M	7500	3	Tasmania
1117	Fauchery	M	8340	1	Tasmania
1120	Woodfull	M	7890	1	Tasmania

Figure 3: Query results for T1CA_5520

B. Task Credit1b

In Credit Task 1b, the task is slightly more challenging because it requires students to create a query with two separate sets of criteria connected by the OR operator. In addition, the query must display the required customer information while filtering records that satisfy either of the specified conditions. Therefore, the Query Design Grid includes Customer ID, Surname, Sales This Year, Sales Last Year, Loyalty Points, Rating Code, and State, together with the criteria provided in the assignment description with following matching criteria:

- Sales this year must be greater than or equal to 560
- The Rating code must be 1
- The state must be Tasmania
- The loyalty points must be in the range 3000 to 5999

OR

- Sales last year must be less than 75
- The Rating code must be 2
- The state must be Victoria
- The loyalty points must be in the range 7500 to 8999

Field:	CustId	Surname	Gender	LoyaltyPts	Rating	State	SalesThisYear	SalesLastYear
Table:	CUSTOMERS520	CUSTOMERS520	CUSTOMERS520	CUSTOMERS520	CUSTOMERS520	CUSTOMERS520	CUSTOMERS520	CUSTOMERS520
Sort:								
Show:	☒	☒	☒	☒	☒	☒	☒	☒
Criteria:				Between 3000 And 5999 1		"Tasmania"	>=560	
or:				Between 7500 And 8999 2		"Victoria"		<75

Figure 4: Query Design Grid for T1CB_5520 Showing Multiple Criteria with OR Conditions

CustId	Surname	Gender	LoyaltyPts	Rating	State	SalesThisYea	SalesLastYea
1007	Foskett	M	5100		1 Tasmania	758	114
1059	Dunningham	M	3200		1 Tasmania	601	205
1061	Melocco	F	3620		1 Tasmania	657	263
1062	Cornish	M	4770		1 Tasmania	989	20
1064	Bungaree	M	4470		1 Tasmania	1012	1873
1071	Ranken	F	3980		1 Tasmania	762	564
1140	Dobie	M	5090		1 Tasmania	700	651
1174	Buckland	F	7560		2 Victoria	0	0
1202	Benjafield	F	8800		2 Victoria	159	55
1226	Nina	M	8820		2 Victoria	238	60
1269	Unaipon	F	8200		2 Victoria	0	0
1313	Opas	M	8650		2 Victoria	30	21
1341	Lush	F	8200		2 Victoria	10	12
1346	Jardine	M	7770		2 Victoria	7	4
1347	Perry	F	8250		2 Victoria	65	15
1455	Hoffnung	F	8660		2 Victoria	0	0
1490	Ash	F	8910		2 Victoria	13	19
1494	Tewksbury	F	8210		2 Victoria	0	0
1753	Anthon	M	8150		2 Victoria	0	0

Figure 5: Query Results for T1CB_5520 Showing Customers Matching Alternative Criteria

C. Task Credit1c

In the third task, the lab requires students to generate query named *T1CC_5520* contains following attributes Customer ID, given name, surname, date of birth and postcode. However, the condition of DateOfBirth requires within years 31st Dec 1999 to 31st Dec 2001

Field:	CustId	GivenName	Surname	DateOfBirth	PostCode
Table:	CUSTOMER5520	CUSTOMER5520	CUSTOMER5520	CUSTOMER5520	CUSTOMER5520
Sort:					
Show:	☒	☒	☒	☒	☒
Criteria:				Between #12/31/1999#	
or:					

Figure 6: Query Design Grid for T1CC_5520 Showing Customers by Date of Birth Range

CustId	GivenName	Surname	DateOfBirth	PostCode
1002	Audrey	Ngabidj	6/7/2000	7150
1079	Piper	Lazarev	4/8/2001	7250
1522	Olivia	Medley	1/5/2000	3377
1762	Aiden	Salisbury	7/4/2000	3636
1999	Charlotte	Glassey	7/3/2000	5062
2366	Ashley	Pearce	5/2/2000	2715
2375	Mia	Douglass	6/6/2000	2140
2682	Anna	Cracknell	10/4/2000	2325
2965	Logan	Harney	2/4/2000	2439
3037	Logan	Kohler	6/13/2000	2440
3341	Alyssa	Bogle	5/12/2000	4275
3435	Abbey	Kayser	3/26/2000	4303
3603	Jade	Whitlam	4/22/2000	4427
3900	Justin	Oakes	9/8/2000	4856

Figure 7: Query Results for T1CC_5520 (Customer Date of Birth Filter)

D. Task Credit1d

In the fourth task, students are required to add their own information as a new record in the CUSTOMER5520 table.

4001 M	Mr.	Hieu	Truong	Bach Dang	Ho Chi Minh	Tan Son Hoa	4872 VietNam	5000	2500	6000	1	-24.32	145.32	9/8/21
Mr.	Hieu	Truong	Bach Dang	Ho Chi Minh	Tan Son Hoa	4872 VietNam	5000	2500	6000	1	-24.32	145.32	9/8/2006	9/8/2024

Figure 8: Newly Added Customer Record in the Customer Table.

E. Task Credit1e

Before setting Relationships, student has to download file **ProductSales DB [ZIP, 11.5MB]** on Canvas and rename all tables by adding last 4 digits of your student ID at the end of each table name.

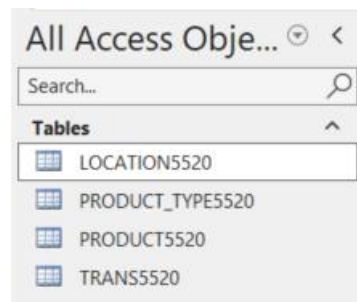


Figure 9: Renamed Product Sales Database Tables with Student ID Suffix

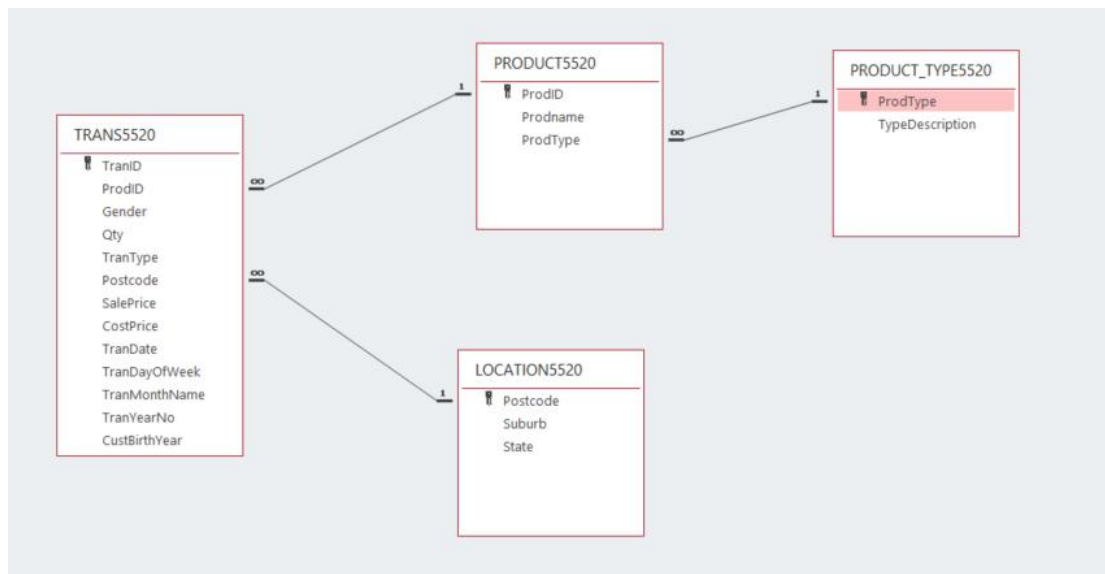


Figure 10: Table Relationships Diagram for the Product Sales Database

It is clear that TRANS5520.Postcode is linked to LOCATION5520.Postcode, TRANS5520.ProdID is linked to PRODUCT5520.ProdID, and PRODUCT5520.ProdType is linked to PRODUCT_TYPE5520.ProdType.

F. Task Credit1f

In Task Credit1f, the Lab asks students to create query named **T1CF_4567** that displays these columns: TranID, ProdID, Prodname, TypeDescription, Qty, TranDate in Ascending TransID sequence. Additionally, Qty greater than or equal to 10 and TranDate = 01/03/2011

Field:	TranID	ProdID	Prodname	TypeDescription	Qty	TranDate
Table:	TRANS5520	TRANS5520	PRODUCT5520	PRODUCT_TYPE5520	TRANS5520	TRANS5520
Sort:	Ascending					
Show:	☒	☒	☒	☒	☒	☒
Criteria:					>=10	#1/3/2011#
or:						

Figure 11: Query Design for T1CF_5520

TranID	ProdID	Prodname	TypeDescript	Qty	TranDate
7786824	10	Sanstrong	LifeStyle	10	1/3/2011
8150903	66	Techiflex	Fashion	24	1/3/2011
24340777	2	Domapbam	LifeStyle	12	1/3/2011
26196643	28	Viaholdings	Pharmacy	10	1/3/2011

Figure 12: Query Results for T1CF_5520 (Datasheet View)

G. Task Credit1g

Subsequently, the taskCreditg asks student to create query named **T1CG_5520** that displays these columns: TranDate, TranID, Prodname, SalePrice, Qty, Suburb, CustBirthYear:

- Ascending TransDate sequence
- Prodname is "Highway" or "Touchtrax "
- Suburb is Balwyn

Field:	TranDate	TranID	Prodname	SalePrice	Qty	Suburb	CustBirthYear
Table:	TRANS5520	TRANS5520	PRODUCT5520	TRANS5520	TRANS5520	LOCATION5520	TRANS5520
Sort:	Ascending						
Show:	☒	☒	☒	☒	☒	☒	☒
Criteria:			"Highway" Or "Touchtra			"Balwyn"	
or:							

Figure 13: Query Design for T1CG_5520 (Product and Suburb Filtering)

TranDate	TranID	Prodname	SalePrice	Qty	Suburb	CustBirthYear
6/11/2010	53024430	Touchtrax	\$42.50	20	BALWYN	1959
3/10/2011	58413842	Highway	\$70.40	2	BALWYN	1994
4/13/2011	50058779	Touchtrax	\$44.60	5	BALWYN	1956
4/30/2011	82193951	Touchtrax	\$44.60	2	BALWYN	1962
6/28/2011	55848363	Touchtrax	\$44.60	5	BALWYN	1985
4/13/2012	6666705	Touchtrax	\$46.80	2	BALWYN	1968
5/5/2012	56828278	Touchtrax	\$46.80	10	BALWYN	1951
2/15/2013	72758989	Touchtrax	\$48.90	2	BALWYN	1950
6/9/2013	99698475	Touchtrax	\$48.90	20	BALWYN	1974
6/28/2013	26534460	Highway	\$77.00	2	BALWYN	1990
8/2/2013	23614714	Touchtrax	\$48.90	20	BALWYN	2000
9/4/2013	29939836	Touchtrax	\$48.90	5	BALWYN	2000
5/17/2014	75315658	Touchtrax	\$51.00	5	BALWYN	1960
9/13/2014	80040816	Touchtrax	\$51.00	10	BALWYN	1968
12/4/2014	91587436	Highway	\$80.40	4	BALWYN	1999

Figure 14: Query Results for T1CG_5520 (Datasheet View)

H. Task Credit1h

In this part, the question is divided into two smaller questions for I will answer respectively.

First, for answering how many times has Touchtrax product been sold, I decided to create new aquery name T1CH_01_5520 with following step configurations:

1. Create new Query Design.
2. Choose two tables: PRODUCT5520 and TRAN5520
3. Add Fields to the Grid:
 - Add the Prodname field (from the PRODUCT5520 table).
 - Add the TranID field (from the TRANS5520 table).
4. Enable Totals: On the Design tab, click the Totals (\$\Sigma\$) button to display the Total row in the design grid.
5. Configure Criteria and Functions:
 - In the Prodname column, set the Total row to Group By and enter "Touchtrax" in the Criteria row.
 - In the TranID column, set the Total row to Count.
6. Run the Query

Field:	Prodname	TranID
Table:	PRODUCT5520	TRANS5520
Total:	Group By	Count
Sort:		
Show:	☒	☒
Criteria:	"Touchtrax"	
or:		

Figure 15: Query Design for Counting Touchtrax Sales

T1CH-01_5520 X	
Prodname	CountOfTran
Touchtrax	4725

Figure 16: Query Result for Counting Touchtrax Sales (Datasheet View)

After configuring and running the query, it shows that the product Touchtrax has been sold 4,725 times.

Then, for answering question that How many different prices has this product been sold for, the following descriptions showcase how can I do it

1. Create Query Design: Open Access, go to the Create tab, and click Query Design.
2. Select Tables: Add the PRODUCT5520 and TRANS5520 tables to the design window.
3. Add Fields to the Grid:
 - Add the Prodname field (from the PRODUCT5520 table).
 - Add the SalePrice field (from the TRANS5520 table).
4. Enable Totals: On the Design tab, click the Totals (Σ) button to display the Total row in the design grid.
5. Configure Criteria and Functions:
 - In the Prodname column, set the Total row to Group By and enter "Touchtrax" in the Criteria row.
 - In the SalePrice column, set the Total row to Group By to list all unique selling prices without duplication.

6. Run the Query

Field:	Prodname	SalePrice
Table:	PRODUCT5520	TRANS5520
Total:	Group By	Group By
Sort:		
Show:	☒	☒
Criteria:	"Touchtrax"	
or:		

Figure 17: Query Design for Distinct Selling Prices

Prodname	SalePrice
Touchtrax	\$42.50
Touchtrax	\$44.60
Touchtrax	\$46.80
Touchtrax	\$48.90
Touchtrax	\$51.00

Figure 18: Query Design for Distinct Selling Prices (Datasheet View)

Based on the datasheet view, there are a total of five different prices for the product Touchtrax.

Subsequently, I create a new query named **T1CH-03_5520**, select tables **PRODUCT5520** and **TRANS5520**, add fields including **Prodname**, **TranDate**, **SalePrice**,. Then, configure column **Prodname** enter “Touchtrax” in **Criteria** and **Ascending** in TranDate.

Field:	Prodname	TranDate	SalePrice
Table:	PRODUCT5520	TRANS5520	TRANS5520
Sort:		Ascending	
Show:	☒	☒	☒
Criteria:	"Touchtrax"		
or:			

Figure 19: Query Design for price trend Analysis

Touchtrax	12/21/2010	\$42.50
Touchtrax	12/22/2010	\$42.50
Touchtrax	12/24/2010	\$42.50
Touchtrax	12/26/2010	\$42.50
Touchtrax	12/28/2010	\$42.50
Touchtrax	12/30/2010	\$42.50
Touchtrax	1/1/2011	\$44.60
Touchtrax	1/2/2011	\$44.60
Touchtrax	1/7/2011	\$44.60
Touchtrax	1/7/2011	\$44.60
Touchtrax	1/13/2011	\$44.60
Touchtrax	1/16/2011	\$44.60

Figure 20: Query Result for Price Trend Analysis (Datasheet View)

Based on the query result, the price of the product remains stable within the same year and increases progressively across different years. In detail, the sale price of the product Touchtrax was \$42.50 in 2010, then increased to \$44.60 the following year, and continued to rise, reaching \$51.00 by 2014.

I. Task Credit1i

In the Task1i, a query named **T1Ci_5520** was created using the **TRANS5520** table to calculate the sum of the quantity sold for each weekday, including Saturday and Sunday. In the query design, a calculated expression using weekday formatting was set to Group By, while the Qty field was configured with the Sum function to

aggregate the total sales volume. Additionally, the Qty column was sorted in Ascending order to arrange the list from the lowest to the highest total quantity. The resulting datasheet view successfully displays each weekday alongside its corresponding total quantity sold, ordered sequentially as required (*WeekdayName(Weekday([TranDate]))*).

Field:	Expr1: WeekdayName	Qty
Table:		TRANS5520
Total:	Group By	Sum
Sort:		Ascending
Show:	☒	☒
Criteria:		
or:		

Figure 21: Query Design for T1Ci_5520 (Weekday Sales Quantity Sum)

Expr1	SumOfQty
Monday	159615
Friday	159919
Sunday	160359
Wednesday	160640
Thursday	160693
Tuesday	161288
Saturday	163064

Figure 22: Query result for T1Ci_5520 (Datasheet View)

J. Task Credit1j

PRODUCT5520

PRODUCT5520

Form Header

Student Name: Truong Ngoc Gia Hieu

Student ID: SWS01217/105565520

Photo

Detail

ProdID

Prodname

ProdType

Photo

Table.TRANS5520

Form Footer

Figure 23: Form Design View for PRODUCT5520

PRODUCT5520 X

Student Name: Truong Ngoc Gia Hieu
Student ID: SWS01217/105565520



ProdID:

Prodname:

ProdType:

Photo:

TranID	Gender	Qty	TranType	Postcode	SalePrice	CostPrice	TranDate	Tr
10502	Female	12	Sale	3075	\$53.00	\$27.90	7/5/2011 We	
36135	Female	2	Sale	2615	\$60.60	\$31.90	7/19/2014 Sun	
56017	Female	6	Sale	2173	\$50.50	\$26.60	6/13/2010 Mo	
89472	Female	6	Sale	2460	\$58.10	\$30.60	5/22/2013 Thu	
140895	Female	12	Sale	2650	\$55.60	\$29.30	8/6/2012 Tue	
146349	Female	6	Sale	2603	\$50.50	\$26.60	7/9/2010 Sat	
277039	Female	12	Sale	3199	\$60.60	\$31.90	7/18/2014 Sat	
307332	Female	3	Sale	5171	\$53.00	\$27.90	9/4/2011 Mo	

Record: 1 of 1177 No Filter Search

Figure 24: Form View for PRODUCT5520

CONCLUSION

Through this lab, I mastered essential database management skills in Microsoft Access, including applying standardized naming conventions (_5520), building advanced analytical queries using criteria, aggregation (Sum), and date functions (WeekdayName), and designing professional forms with customized image formatting (SizeMode) and embedded transaction subforms.